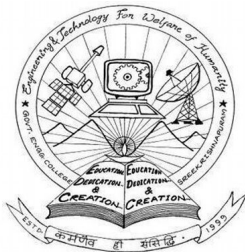


ITT464 COMPUTER GRAPHICS

Module 3 : 2-Dimensional Geometric Transformations



Anoop S K M

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Govt. Engg. College, Sreekrishnapuram, Palakkad

Acknowledgements

- Some pictures are taken from the Internet using Google search.
- Some are taken from the book Computer Graphics by Hearn & Baker
- Wikipedia also referred.

What's in Store?

Syllabus

Module 1: Graphics Systems (8 Hours)
Raster Scan & Random Scan systems. Output Primitives – Line Drawing Algorithms (DDA, Bresenham), Bresenham's circle generation algorithm. Filled Area Primitives – Scan Fill, Flood Fill, Boundary Fill. Inside outside tests.
Module 2: Display Technologies(6 Hours)
Working principle behind CRT, LCD, Plasma, LED, OLED, AMOLED, E-Paper displays.
Module 3: 2-Dimensional Geometric Transformations (8 Hours)
Basic Transformations, Reflection & Shear, Homogenous Matrix representation of transformations, Composite Transformations.
Module 4: 2-D Clipping & 3-Dimensional Geometric Transformations (6 Hours)
2-D Clipping: Point Clipping, Cohen-Sutherland Line Clipping Algorithm, Sutherland-Hodgeman Polygon Clipping Algorithm. 3-Dimensional Geometric Transformations: Basic transformations, Composite 3D transformations (Basics only)
Module 5: Visible Surface Detection Methods & Colour Models (7 Hours)
Visible Surface Detection Methods: Back Face Detection, Depth Buffer, A-Buffer, Scan line, Depth sorting methods. Colour Models: RGB Model, CMY Model, HSV Model

Text Books

1. Donald Hearn, Pauline Baker, Computer Graphics – C Version, Pearson Education, 2nd Edition, 2002
2. Amarendra N' Sinha and Arun D Udai, "Computer Graphics", The McGraw-Hill

Meeting 07

We Saw & Will See

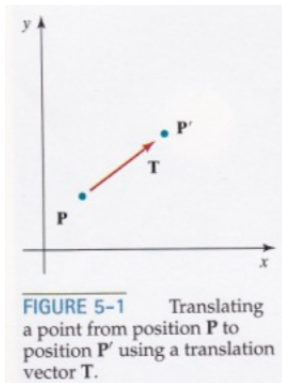
Till Now We Saw...

- Module 1 : Graphic Systems
 - Raster Scan & Random Scan
 - Line Drawing Algorithm : DDA, Bresenham's
 - Bresenham's Circle(Midpoint Circle)
 - Filling Algorithms: Scan Line Fill , Flood Fill , Boundary Fill
 - Inside-Outside Tests : Odd-Even, Non-zero Winding

Today We Will See...

- Module 3:
 - Basic Transformations : Translation, Rotation and Scaling

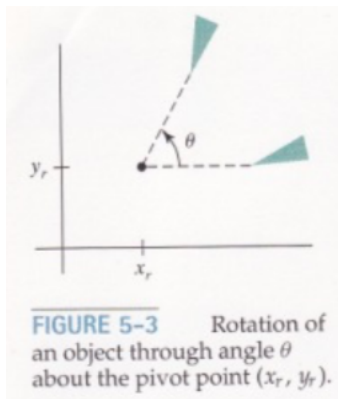
Translation



$$P' = P + T$$

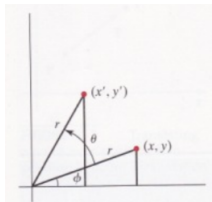
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

Rotation



- rotation by an angle Θ in anti-clockwise

Rotation Matrix



$$x = r \cos \phi, y = r \sin \phi$$

$$x' = r \cos(\phi + \Theta)$$

$$= r \cos \phi * \cos \Theta - r \sin \phi * \sin \Theta$$

$$= x \cos \Theta - y \sin \Theta$$

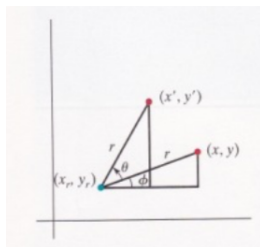
$$y' = r \sin(\phi + \Theta)$$

$$= r \cos \phi * \sin \Theta + r \sin \phi * \cos \Theta$$

$$= x \sin \Theta + y \cos \Theta$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \Theta & -\sin \Theta \\ \sin \Theta & \cos \Theta \end{bmatrix} * \begin{bmatrix} x \\ y \end{bmatrix}$$

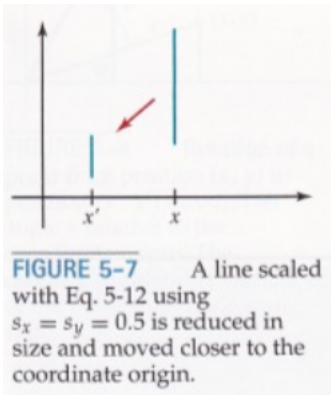
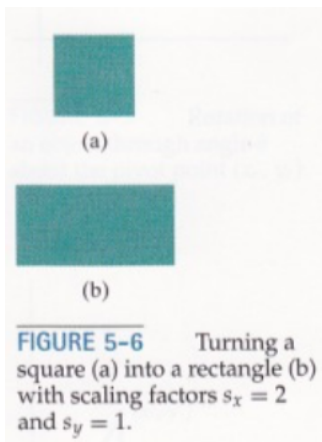
Rotation with respect to (x_c, y_c)



$$x' = x_c + (x - x_c) \cos \Theta - (y - y_c) \sin \Theta$$

$$y' = y_c + (x - x_c) \sin \Theta + (y - y_c) \cos \Theta$$

Scaling



Scaling Matrix

Scaling with respect to origin by S_x, S_y along x, y axes respectively.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} * \begin{bmatrix} x \\ y \end{bmatrix}$$

Scaling with respect to (x_c, y_c) by S_x, S_y along x, y axes respectively.

$$x' = x_c + (x - x_c) * S_x$$

$$y' = y_c + (y - y_c) * S_y$$

Two-dimensional Translation Matrix

The equation for translation is

$$x' = x + t_x, y' = y + t_y$$

Its corresponding matrix representation is

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Two-dimensional Rotation Matrix

The equation for rotation is

$$x' = x \cos \Theta - y \sin \Theta, y' = x \sin \Theta + y \cos \Theta$$

Its corresponding matrix representation is

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \Theta & -\sin \Theta & 0 \\ \sin \Theta & \cos \Theta & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Two-dimensional Scaling Matrix

The equation for translation is

$$x' = x * S_x, y' = y * S_y$$

Its corresponding matrix representation is

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Meeting 08

We Saw & Will See

Till Now We Saw...

- Module 1 : Graphic Systems
 - Raster Scan & Random Scan
 - Line Drawing Algorithm : DDA, Bresenham's
 - Bresenham's Circle(Midpoint Circle)
 - Filling Algorithms: Scan Line Fill , Flood Fill , Boundary Fill
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- Module 3:
 - Basic Transformations : Translation, Rotation and Scaling

Today We Will See...

- Composite Transformations

Two-dimensional Composite Translation Matrix

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_{x_2} \\ 0 & 1 & t_{y_2} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & t_{x_1} \\ 0 & 1 & t_{y_1} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$
$$= \begin{bmatrix} 1 & 0 & t_{x_1} + t_{x_2} \\ 0 & 1 & t_{y_1} + t_{y_2} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Two-dimensional Composite Rotation Matrix

$$\begin{aligned}
 \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} &= \begin{bmatrix} \cos \Theta_2 & -\sin \Theta_2 & 0 \\ \sin \Theta_2 & \cos \Theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \Theta_1 & -\sin \Theta_1 & 0 \\ \sin \Theta_1 & \cos \Theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \\
 &= \begin{bmatrix} \cos \Theta_1 \cos \Theta_2 - \sin \Theta_1 \sin \Theta_2 & -\sin \Theta_1 \cos \Theta_2 - \cos \Theta_1 \sin \Theta_2 & 0 \\ \cos \Theta_1 \sin \Theta_2 + \sin \Theta_1 \cos \Theta_2 & \cos \Theta_1 \cos \Theta_2 - \sin \Theta_1 \sin \Theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \\
 &= \begin{bmatrix} \cos(\Theta_1 + \Theta_2) & -\sin(\Theta_1 + \Theta_2) & 0 \\ \sin(\Theta_1 + \Theta_2) & \cos(\Theta_1 + \Theta_2) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}
 \end{aligned}$$

Two-dimensional Composite Scaling Matrix

$$\begin{aligned}\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} &= \begin{bmatrix} S_{x_2} & 0 & 0 \\ 0 & S_{y_2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} S_{x_1} & 0 & 0 \\ 0 & S_{y_1} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} S_{x_1} S_{x_2} & 0 & 0 \\ 0 & S_{y_1} S_{y_2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}\end{aligned}$$

Meeting 08

We Saw & Will See

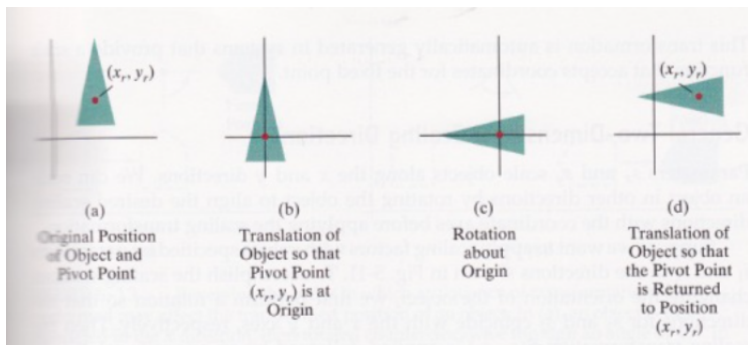
Till Now We Saw...

- Module 1 : Graphic Systems
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- Module 3:
 - Basic Transformations : Translation, Rotation and Scaling
 - Homogeneous Matrix Representations
 - Composite Transformations

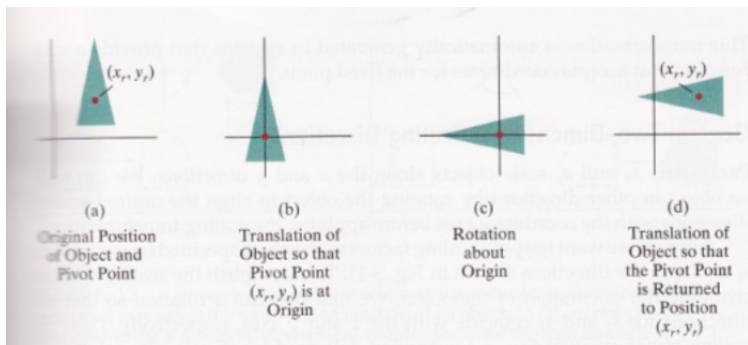
Today We Will See...

- More on Composite 2-D Transformation

Composite 2-D Transformation- Pivot Point Rotation

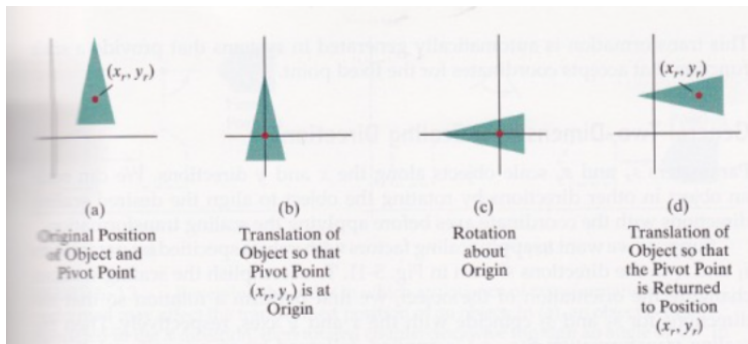


Composite 2-D Transformation- Pivot Point Rotation



$$R(x_f, y_f, \Theta) = T(x_f, y_f)R(\Theta)T(-x_f, -y_f)$$

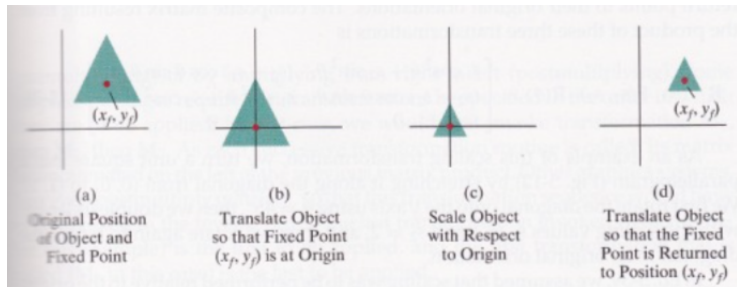
Composite 2-D Transformation- Pivot Point Rotation



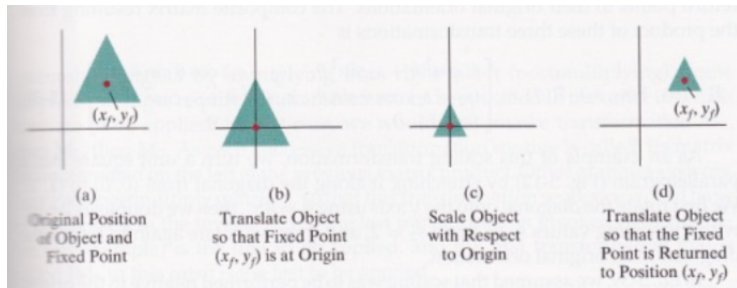
$$R(x_f, y_f, \Theta) = T(x_f, y_f)R(\Theta)T(-x_f, -y_f)$$

Find the Transformation Matrix

Composite 2-D Transformation- Pivot Point Scaling

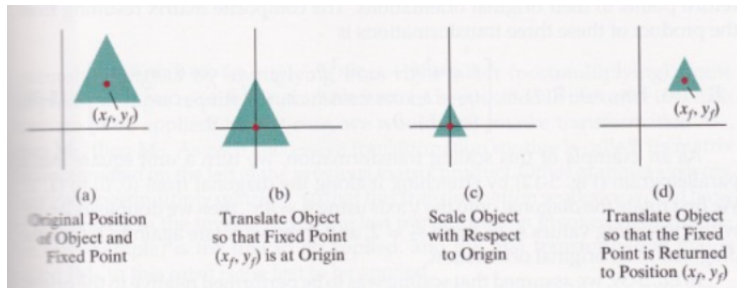


Composite 2-D Transformation- Pivot Point Scaling



$$S(x_f, y_f, S_x, S_y) = T(x_f, y_f)S(S_x, S_y)T(-x_f, -y_f)$$

Composite 2-D Transformation- Pivot Point Scaling



$$S(x_f, y_f, S_x, S_y) = T(x_f, y_f)S(S_x, S_y)T(-x_f, -y_f)$$

Find the Transformation Matrix

Meeting 09

We Saw & Will See

Till Now We Saw...

- Module 1 : Graphic Systems
 - Raster Scan & Random Scan
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 - Homogeneous Matrix Representations
 - Composite Transformations

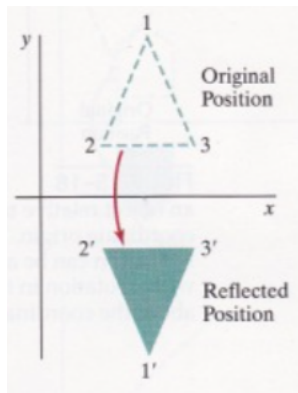
Today We Will See...

- Reflection and Shear

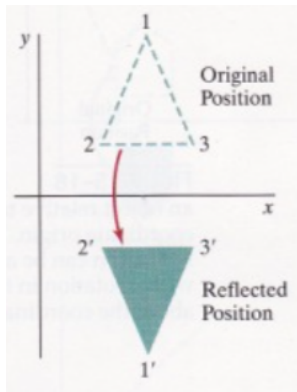
Reflection

- Reflection in 2D is rotation by 180° with respect to an axis

Reflection with respect to x axis



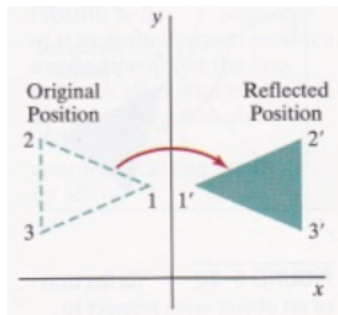
Reflection with respect to x axis



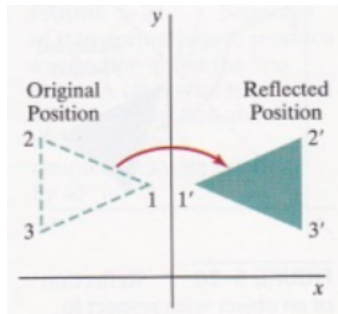
- Reflection of (x, y) with x axis :
 $x' = x$ and $y' = -y$
- Transformation Matrix is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Reflection with respect to y axis



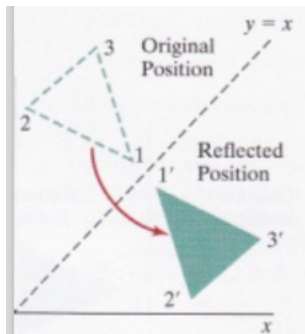
Reflection with respect to y axis



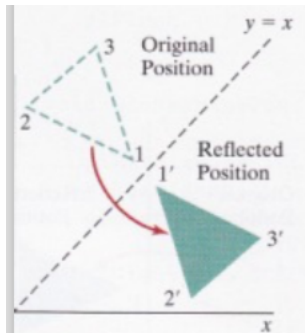
- Reflection of (x, y) with y axis :
 $x' = -x$ and $y' = y$
- Transformation Matrix is

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Reflection with respect to line $x = y$

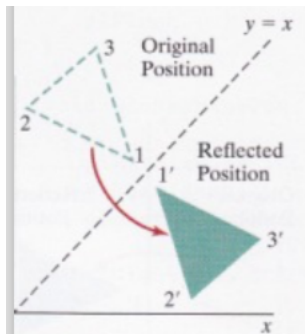


Reflection with respect to line $x = y$



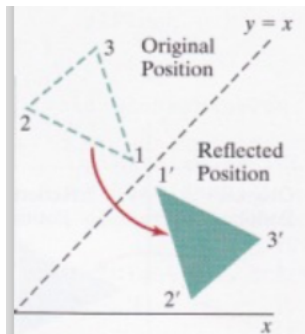
- Reflection of (x, y) with $x = y$ line :

Reflection with respect to line $x = y$



- Reflection of (x, y) with $x = y$ line :
 $x' = y$ and $y' = x$
- Transformation Matrix is

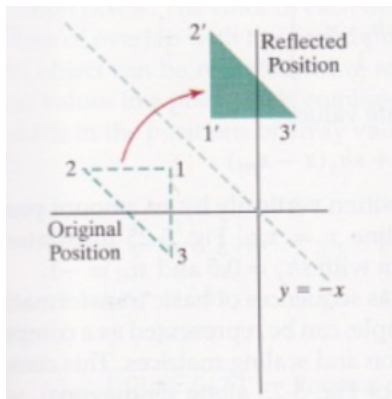
Reflection with respect to line $x = y$



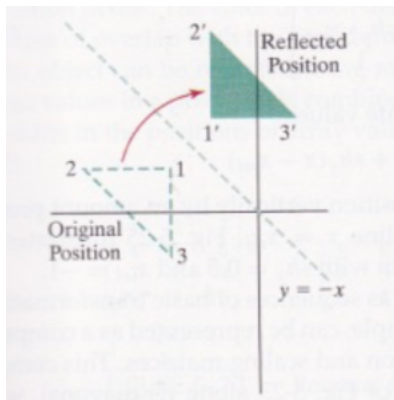
- Reflection of (x, y) with $x = y$ line :
 $x' = y$ and $y' = x$
- Transformation Matrix is

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Reflection with respect to line $x = -y$

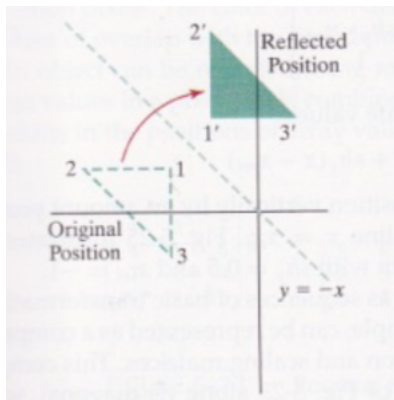


Reflection with respect to line $x = -y$



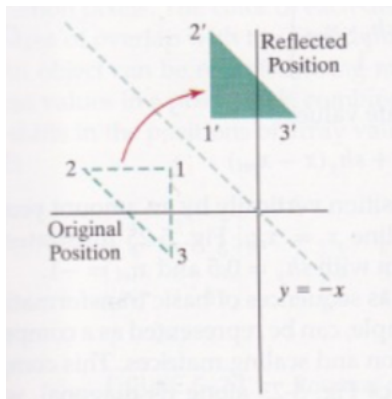
- Reflection of (x, y) with line $x = -y$:

Reflection with respect to line $x = -y$



- Reflection of (x, y) with line $x = -y$:
 $x' = -y$ and $y' = -x$
- Transformation Matrix is

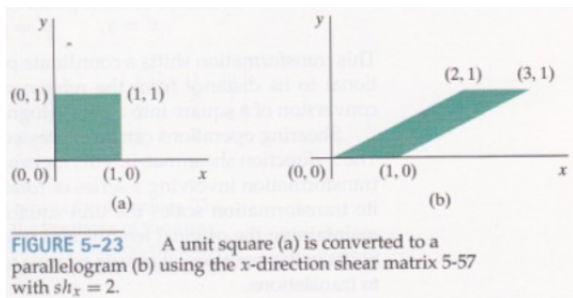
Reflection with respect to line $x = -y$



- Reflection of (x, y) with line $x = -y$:
 $x' = -y$ and $y' = -x$
- Transformation Matrix is

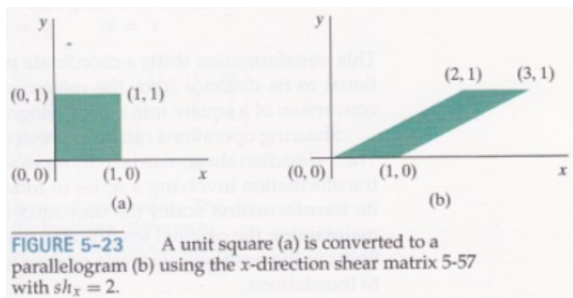
$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Shear



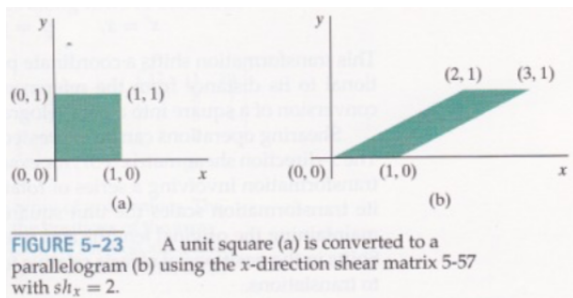
- a transformation that distorts the shape of object

Shear



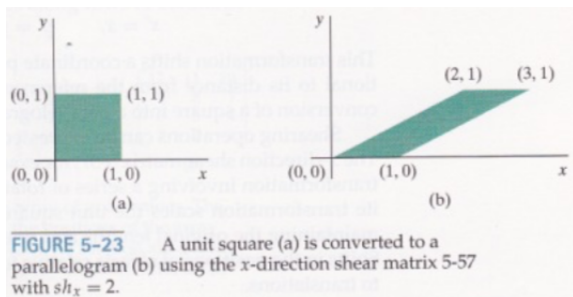
- a transformation that distorts the shape of object
- transformed shape appears as though the object were composed of layers.

Shear



- a transformation that distorts the shape of object
- transformed shape appears as though the object were composed of layers.
- x axis shear $x' = x + sh_x * y, y' = y$

Shear



- a transformation that distorts the shape of object
- transformed shape appears as though the object were composed of layers.
- x axis shear $x' = x + sh_x * y, y' = y$
- y axis shear $x' = x, y' = y + sh_y * x$

Shear Transformation Matrices

x axis shear : $x' = x + sh_x * y, y' = y$

Shear Transformation Matrices

x axis shear : $x' = x + sh_x * y, y' = y$

$$\begin{bmatrix} 1 & sh_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Shear Transformation Matrices

x axis shear : $x' = x + sh_x * y, y' = y$

$$\begin{bmatrix} 1 & sh_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

y axis shear : $x' = x, y' = y + sh_y * x$

Shear Transformation Matrices

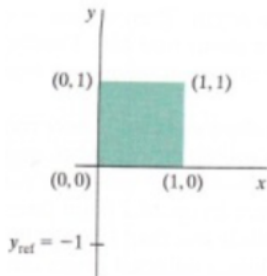
x axis shear : $x' = x + sh_x * y, y' = y$

$$\begin{bmatrix} 1 & sh_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

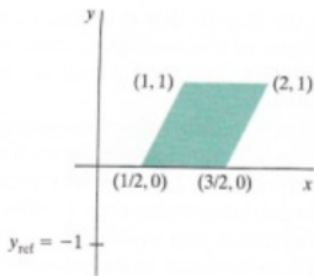
y axis shear : $x' = x, y' = y + sh_y * x$

$$\begin{bmatrix} 1 & 0 & 0 \\ sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

x-axis Shear with respect to a Reference Point

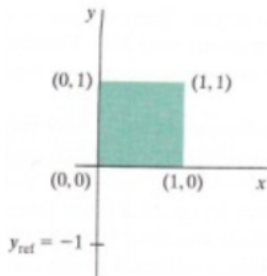


(a)

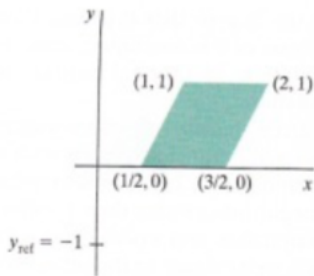


(b)

x-axis Shear with respect to a Reference Point



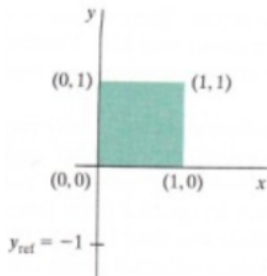
(a)



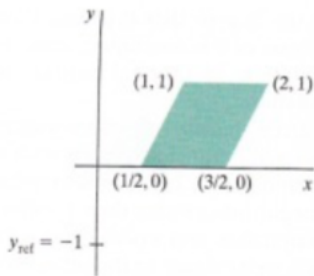
(b)

x axis shear with respect to y_{ref} : $x' = x + (y - y_{\text{ref}})sh_x, y' = y$

x-axis Shear with respect to a Reference Point



(a)

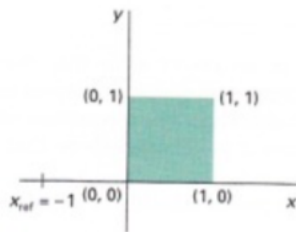


(b)

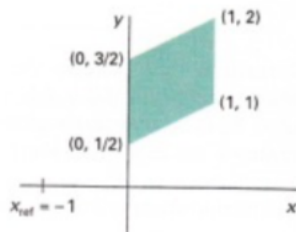
x axis shear with respect to y_{ref} : $x' = x + (y - y_{ref})sh_x$, $y' = y$

$$\begin{bmatrix} 1 & sh_x & -sh_x * y_{ref} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

y-axis Shear with respect to a Reference Point

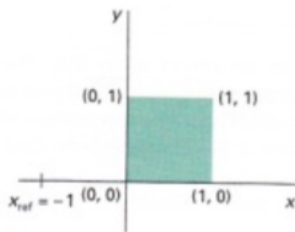


(a)

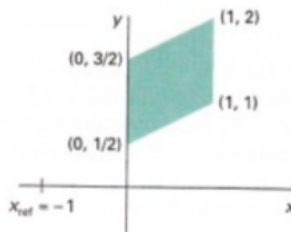


(b)

y-axis Shear with respect to a Reference Point



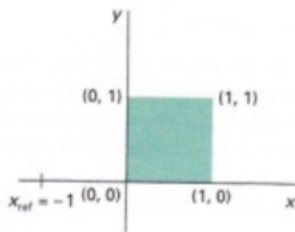
(a)



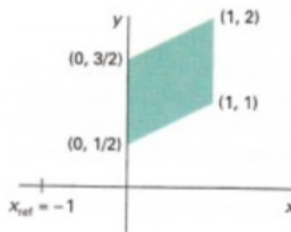
(b)

y axis shear with respect to x_{ref} : $x' = x, y' = y + (x - x_{ref})sh_y$

y-axis Shear with respect to a Reference Point



(a)



(b)

y axis shear with respect to x_{ref} : $x' = x, y' = y + (x - x_{ref})sh_y$

$$\begin{bmatrix} 1 & 0 & 0 \\ sh_y & 1 & -sh_y * x_{ref} \\ 0 & 0 & 1 \end{bmatrix}$$